

Salinization and crop transpiration of irrigated fields in Pakistan's Punjab

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Abstract

In Punjab, Pakistan, scarcity of canal water has lead to a rapid growth of the number of tube wells, pumping poor quality groundwater. In recent years, concerns have been expressed about the threat of salinization through the increasing use of tube well water for irrigation purposes. This paper reviews research results of a model study on the impact of irrigation practices, i.e., irrigation quantity, quality and frequency, on soil salinity and crop transpiration of cotton and wheat in a conjunctive use environment. Four fields with a different soil texture and irrigation water quality were selected and monitored during three cropping seasons. The agro-hydrological model SWAP93 was used to analyse the water flow and salt transport for the prevailing conditions. In the calibration process, the simulated water and salt balance appear to be sensitive to rooting depth, crop factors, actual soil evaporation and the soil retention function. Two concepts were used to account for field-scale heterogeneity of water infiltration and preferential flow. Taking into account this heterogeneity, the percolation losses increase, while the leaching of salts decreases. The calibrated model was used to evaluate the effects of various irrigation practices on salinization and crop transpiration. The irrigation water quantity, in terms of under-irrigation, has a greater impact than the quality of the irrigation water. Irrigation frequency may have a significant impact on the water and salt balance, especially on sandy soils. Finally, long term effects of current

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irrigation practices are discussed, in relation to the different soils in the study area. Soil texture appears to affect crop transpiration and salinization considerably. © 1997 Elsevier Science B.V.

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1. Introduction

Salinity has long been present in the Indus Basin in Pakistan due to the aridity of the climate and weathering of the marine parent material. Lands affected by genetic salinity were generally excluded from the large-scale irrigation systems that have been constructed from the second half of the 19th century onwards. However, the introduction of these irrigation systems caused a rise in groundwater tables, bringing salts into the root zone through capillary rise. In recent years, concerns have been expressed about the threat of salinization through the use of poor quality groundwater pumped by tube wells (Choudry et al., 1978). This concern is prompted by the fact that at present almost 30% of the irrigation water at the farm gate is derived from groundwater, which has a water quality far inferior to canal water (Nespak/SGI, 1991).

Research and engineering works to prevent land degradation and combat waterlogging and associated salinity, have generally focused on vertical drainage and the prevention of seepage from irrigation canals. In the present context, where groundwater tables are falling in large parts of the country due to overpumping and where the use of groundwater is still increasing, research is urgently needed to quantify the threat of salinization through the use of tube well water for irrigation (Kijne and Kuper, 1995).

Since 1988, the International Irrigation Management Institute (IIMI) has conducted integrated field research on inter-related aspects of irrigation, salinity and agricultural production (IIMI, 1996). This paper reports research results of a study on the impact of irrigation practices, i.e., irrigation quantity, quality and frequency, on soil salinity and crop transpiration. The research was conducted on the fields of the farmers in order to capture the wide range of irrigation practices that are employed by farmers in response to the physical environment with which they are faced. The agro-hydrological model SWAP93 (Van den Broek et al., 1994) is used as a tool to study the marginal effects of different irrigation practices on soil salinization and crop transpiration, and to evaluate the long term impact of the current irrigation practices in order to suggest an improved irrigation management.

Currently, research efforts are focused on the Chishtian subdivision in southeast Punjab, a 70,000 ha irrigation scheme, receiving surface irrigation water through the Fordwah Branch canal. Increased cropping intensities have prompted farmers to augment available canal water supplies by groundwater, pumped by over 4000 tube wells in the area (about 7 per 100 ha). Consequently, groundwater tables have dropped and waterlogging is generally not a problem in the area. The average electrical conductivity (EC) of all the tube wells is 0.89 dS/m, with 90% of the tube wells showing an EC between 0 and 4 dS/m. The study area is located in the arid agro-ecological zone, with cotton and wheat as the dominant crops during summer (June till December), and winter season (January till May), respectively. The soils in the area are of alluvial and eolian origin, resulting in a mixture of silty to sandy deposits.

SWAP93 is successfully calibrated and validated, and a sensitivity analysis is carried out to identify the most sensitive input parameters and advice on a suitable data collection. On a short term basis, the marginal impact of so-called management variables, like quantity, quality and frequency of irrigation is assessed. Since the same irrigation management has a different impact on soil salinity and crop transpiration under different environmental conditions, the interaction between irrigation practices and so-called environmental parameters, like soil type and groundwater table is studied. Finally, the calibrated model is used to estimate trends in soil salinity and identify risks for crop transpiration under current irrigation practices in the Chishtian subdivision.

2. Methodology

2.1. Data collection

Four farmers' fields of 0.4 ha each were monitored during three growing seasons in order to collect an extensive data set to calibrate and validate the model SWAP93 (cotton–wheat–cotton cycle from July '94 up to December '95). The four fields represent the predominant soils in the study area, ranging from a loamy sand to a silty clay loam. In Table 1, the soil type and the availability of canal water and/or tube well water is indicated for the four fields. The farmers of field 1 and 2 have access to canal water and both own a tube well to supplement their canal water supplies. The farmers of field 3 and 4 do not have access to canal water, but have a shared ownership of a tube well. The water quality of canal water is excellent with an EC of 0.2 dS/m. The farmers of field 1 and 2 use groundwater from various tube wells with an EC ranging from 0.8 to 1.6 dS/m. The farmers of field 3 and 4 use only water from one tube well, whose quality ranges during the season from 0.75 to 1.3 dS/m. Irrigation regime and crop development was registered by IIMI field staff. Farmers irrigate their bunded fields by flooding, where small cotton beds and furrows are made. Wheat is grown in basins. In each field tensiometers were installed at eight depths (at 15, 30, 45, 60, 90, 120, 150 and 200 cm) and were read every other day. At the beginning of each growing season, ten soil samples are taken at different depths in each field and analysed in the laboratory to determine the soil texture, moisture content and the EC of the saturated extract, EC_e . In the surrounding area of the sample fields, piezometers were frequently read and used to determine groundwater table depths of the four fields. Meteorological data were derived

Table 1
Characteristics of sample fields: soil type, origin and quantity of irrigation water applied during data collection period

Field	Soil type	% Canal water	% Tube well water	Depth irrigation water (cm)
1	Loamy sand (LS)	75	25	396
2	Sandy loam (SL)	67	33	167
3	Loam to silty clay loam (L to SiCL)	—	100	159
4	Loam to silty loam (L to SiL)	—	100	150

from the nearest meteorological stations and precipitation was recorded with rain gauges near the sample fields. Precipitation is about 150–200 mm annually.

2.2. Performance indicators

In this paper two performance indicators are used to evaluate the impact of irrigation management variables and environmental parameters on soil salinity and crop transpiration. These performance indicators relate to a certain time span, for instance a growing season or an agricultural year. The first indicator is relative transpiration RT:

$$RT = \frac{T_{\text{act}}}{T_{\text{pot}}} \quad (1)$$

where T_{act} is the actual crop transpiration and T_{pot} the potential crop transpiration (cm). The second indicator is the salt storage change SSC:

$$SSC = \frac{\Delta S}{S_0} \quad (2)$$

where ΔS is the change in salt storage of the soil profile over a certain time span (g/m^2) and S_0 the initial salt storage of the soil profile (g/m^2).

2.3. The model SWAP93

Feddes et al. (1978) developed the one-dimensional model SWATR to describe transient water flow in a heterogeneous soil–root system, which is under the influence of groundwater. This model was further developed by Belmans et al. (1983), Wesseling et al. (1991), Van den Broek et al. (1994) and Van Dam et al. (1997) and is now referred to as SWAP93, aimed at simulating water, solute and heat transport in the air–plant–soil environment. It is written in FORTRAN 77 and can run on 486 or higher IBM compatible PC. SWAP93 now includes hysteresis according to the concept of Kool and Parker (1987), the possibility for simulating preferential flow, and adsorption and decomposition processes as described by Boesten and Van der Linden (1991).

The upper boundary conditions of the system are determined by the potential evapotranspiration, irrigation and precipitation. Potential evaporation is reduced to the actual evaporation by taking the minimum of (1) the potential evaporation rate, (2) the maximum soil water flux in the top soil according to Darcy assuming a minimum allowed pressure head in the atmosphere, or (3) the empirical evaporation rate according to either Black et al. (1969) or Boesten and Stroosnijder (1986). The model of Black calculates the cumulative actual soil evaporation during a drying cycle by multiplying the square root of the time after the last irrigation or rainfall event with a soil dependent empirical factor, while the cumulative actual soil evaporation according to Boesten and Stroosnijder is equal to the square root of the cumulative potential evaporation multiplied by the soil dependent empirical Boesten coefficient. Actual transpiration rate is equal to the root water uptake rate. SWAP93 calculates the soil water flow with the equation of Richards, based on the law of Darcy and the principle of mass conservation.

In order to solve this equation the soil hydraulic functions of each soil layer should be known. The soil hydraulic functions, expressing the relationships between soil moisture content, pressure head and unsaturated hydraulic conductivity, are described by the Van Genuchten–Muallem parameters (VGM) (Van Genuchten, 1987; Wösten and Van Genuchten, 1988). Various researchers have determined VGM-parameter sets for different soil textural classes (Rawls et al., 1982; Carsel and Parrish, 1988; Wösten et al., 1987). The root water uptake is described semi-empirically by a sink term, which is a function of the maximum root water uptake, the soil water pressure head and the salt concentration. The maximum root water uptake may be uniform or linearly declining with depth (Feddes et al., 1988; Prasad, 1988). At the bottom of the system, boundary conditions can be described with various options, e.g., water table depth, flux to groundwater or free drainage (Wesseling et al., 1991).

SWAP93 describes the transport of conservative solutes with the convection–dispersion equation. No distinction is made between different cations and anions and only the total amount of salts is considered. Saline soil moisture conditions reduce the root water uptake due to an increased osmotic head of the soil water. In SWAP93, the osmotic head is added to the matrix head, and the total head is used to derive the reduction factor for root water uptake. SWAP93 reduces crop transpiration due to water and salinity stress only, while all other conditions are assumed to be optimal, e.g., nutrient availability, cultural practices, pest and disease control.

Movement of water in the unsaturated zone is often not uniform with part of the infiltrating water travelling faster than the average wetting front. This phenomenon is generally called preferential flow, and has a large effect on the leaching of salts to the saturated zone. Preferential flow is incorporated in the SWAP93 through the ‘mobile–immobile’ concept (Van Dam et al., 1996). The soil profile is divided in a mobile part, where water movement and root extraction take place, and an immobile part with a constant soil moisture content. The only interaction between the immobile and the mobile fraction is the diffusion of salts driven by concentration gradients.

3. Model calibration

3.1. Procedure

Determination of the input parameters for the calibration of the model is ideally done by measuring them independently. The main input parameters for the calibration are the VGM-parameters, as they have a considerable impact on unsaturated soil water movement and thus, on salt transport. In situ determination of the VGM-parameters is very difficult. For two sample fields, pressure outflow experiments were carried out in order to determine the ranges for the VGM-parameters of these soils (Beekma et al., 1993). Since these ranges were very wide, the practical use of them was limited. The use of available soil characteristics of a standard series is a possibility to avoid time consuming and costly measurements (Bouma and Van Lanen, 1987). In the present study, VGM-parameters, derived from Wösten et al. (1987) were used as initial values for the calibration process (Table 2) and were slightly modified in a ‘trial and error’ process to

Table 2

Model input data after calibration

General								
Boesten parameter	$\beta = 0.90 \text{ cm}^{1/2}$							
Crop factors	Cotton: 0.5–0.8–1.2–0.9–0.7; wheat: 0.4–0.8–1.15–0.7–0.3							
Maximum rooting depth	Cotton: 140 cm; wheat: 110 cm							
Limiting pressure heads	$h_1 = -0.1 \text{ cm}$; $h_2 = -1.0 \text{ cm}$; $h_3 = -500 \text{ cm}$; $h'_3 = -900 \text{ cm}$; $h_4 = -16,000 \text{ cm}$							
Field specific	Field 1		Field 2		Field 3		Field 4	
	Layer 1	Layer 2	Layer 1	Layer 2	Layer 1	Layer 2	Layer 1	Layer 2
Depth of layer (cm)	0–140	140–315	0–125	125–290	0–105	105–210	0–105	105–210
Soil texture	LS	S	SL	LS	SiCL	LS	SiL	LS
Residual moisture content θ_r	0.01	0.02	0.045	0.02	0.05	0.02	0.045	0.02
Saturated moisture content θ_s	0.33	0.35	0.33	0.35	0.39	0.35	0.38	0.35
Saturated hydr. cond. _s (cm/d)	45.0	150.0	40.0	90.0	16.0	90.0	12.0	90.0
Shape parameter α (1/cm)	0.028	0.026	0.050	0.028	0.030	0.028	0.016	0.028
Shape parameter n	2.1	2.6	1.8	2.6	1.6	2.6	1.6	2.6
Shape parameter λ	0.0	1.0	–0.5	1.0	–1.0	1.0	–1.0	1.0
Gr. water table depth (m)	–	2.80	–	2.50	–	free dr.	–	free dr.

obtain optimal calibration results. In this way, simulation results were matched with measured data, such as pressure heads, soil moisture contents and salt concentrations (EC_e). The results of the calibration were evaluated using the graphs of simulated and measured values. Other input data, like groundwater table depths, irrigation depths and meteorological data were kept constant during the calibration process. The reference evapotranspiration was calculated with the CROPWAT model (Smith, 1992) and multiplied with crop factors (Doorenbos and Pruitt, 1977) to derive the potential evapotranspiration. This was then divided over the potential evaporation and the potential transpiration through an empirical formula taking into account the soil cover.

Although the simulated pressure heads and soil moisture profiles corresponded well with the measured data, the simulated EC_e profiles generally showed an underestimation of the measured values. Therefore, the crop factors and the Boesten factor for soil evaporation were slightly increased in order to reduce the percolation and consequently the leaching of salts.

In case of one-dimensional modelling of this more-dimensional problem, we need to consider the heterogeneity of water infiltration over a field. Since the hydraulic conductivity is not homogeneous over a field, more water infiltrates in highly conductive parts, especially in case of basin irrigation. Besides, farmers' non-uniform water distribution over a field contributes considerably to a heterogeneous infiltration, as more water infiltrates close to the field channel than at the end of the field. Heterogeneous infiltration leads to a higher average percolation over a field, while the average leaching of salts is decreased, as the concentration of the percolation water is lower.

In order to account for the heterogeneous infiltration over a field, two concepts were tested to improve calibration results. The first concept incorporates preferential flow in

Table 3

Model input data after calibration for parameters describing preferential flow

Preferential flow parameters	Field 1		Field 2		Field 3		Field 4	
	Layer 1	Layer 2	Layer 1	Layer 2	Layer 1	Layer 2	Layer 1	Layer 2
Immobile moisture content	0.10	0.10	0.10	0.10	0.12	0.06	0.15	0.10
Mobile fraction	0.55	0.55	0.55	0.55	0.60	0.60	0.6	0.65

the model (see Section 2.3). The percolation of water and salts takes place in the mobile phase, while the immobile phase keeps part of the salts in the soil and delays the leaching. The input data used in this concept are listed in Table 3. The second concept divides a field in three rather arbitrary units: 0.375 part, receiving 67% of the irrigation water depth, 0.375 part receiving 100% and 0.25 part, receiving 150%. For conservative variables such as soil moisture content and EC_e , the weighted averages of the simulation results are compared with the measured data.

3.2. Results model calibration

Generally, the simulated pressure heads match quite well with the measured pressure heads. As a typical example, the measured and simulated pressure heads for loamy sand (field 1) are illustrated in Fig. 1. The simulated pressure heads obtained with the preferential flow concept show more pronounced peaks and declines, since the water movement takes place in the mobile phase only, causing a quicker wetting and drainage of the soil profile.

The soil moisture content profiles are satisfyingly simulated by the model for all soil types. In Fig. 2, a typical example is given for silty clay loam (field 3). The width of the

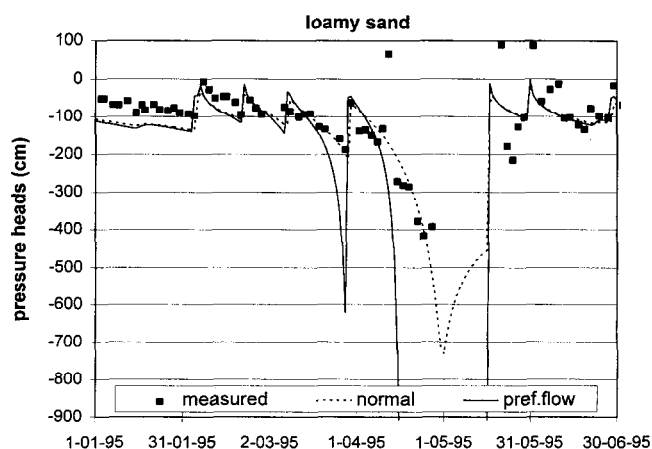


Fig. 1. Measured and simulated pressure heads with normal and preferential flow concept on a loamy sand.

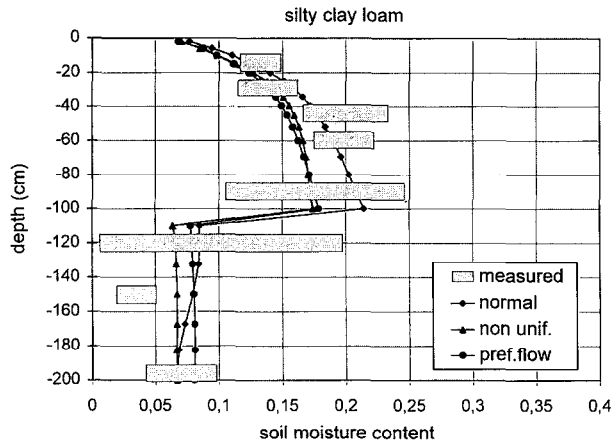


Fig. 2. Measured and simulated moisture content profile with normal, non-uniform, and preferential flow concept on a silty clay loam.

rectangles in Fig. 2 represent the augering inaccuracy of 5 cm depth, while the length of the rectangles represents twice the standard deviation of the measured soil moisture content. In case of the preferential flow concept, less water is stored in the total soil profile, since the soil moisture content in the immobile fraction is relatively low. For the non-uniform concept less water is stored in the under-irrigated part of the field, which cannot be compensated by a higher moisture storage in the over-irrigated part, since field capacity is already reached and the surplus of water percolates.

Fig. 3 illustrates the EC_e profiles in the beginning of July '95 for all soil types. Here, the length of the rectangles represents twice the standard deviation of the measured EC_e . Errors in estimating EC_e can occur due to the dissolution of precipitated salts while saturating the soil sample and thus, increasing the EC_e (Bastiaansen, 1993). Generally, the EC_e profiles of the preferential flow concept are determined by the initial salt content in the immobile phase as there is hardly any interaction between the two fraction (exchange coefficient is set at or close to zero). The salts in the irrigation water are added to the mobile phase only, resulting in a higher percolation and a higher leaching of salts from the mobile phase, while the immobile phase keep the salts in the soil. For the non-uniform concept, accumulation of salts takes place in the under-irrigated part(s) of the field, so the average leaching of salts over the whole field is decreased. Averaging EC_e values over a field, which is done for both the measured values as the non-uniform concept, results in a smoother shape of the EC_e profile.

An evaluation of the three concepts is necessary to decide which one to use in the scenario-analysis. The preferential flow concept has several disadvantages in simulating a heterogeneous water infiltration over a field. The input data for preferential flow concept are determined during the calibration process and, in this context, lack a clear physical meaning. Besides, the initial conditions influence the development of soil salinity to a large extent, especially when the exchange coefficient is set at zero. Thirdly, reduction in transpiration due to salinity stress is not addressed properly, since the salts stored in the immobile fraction do not influence the root water uptake.

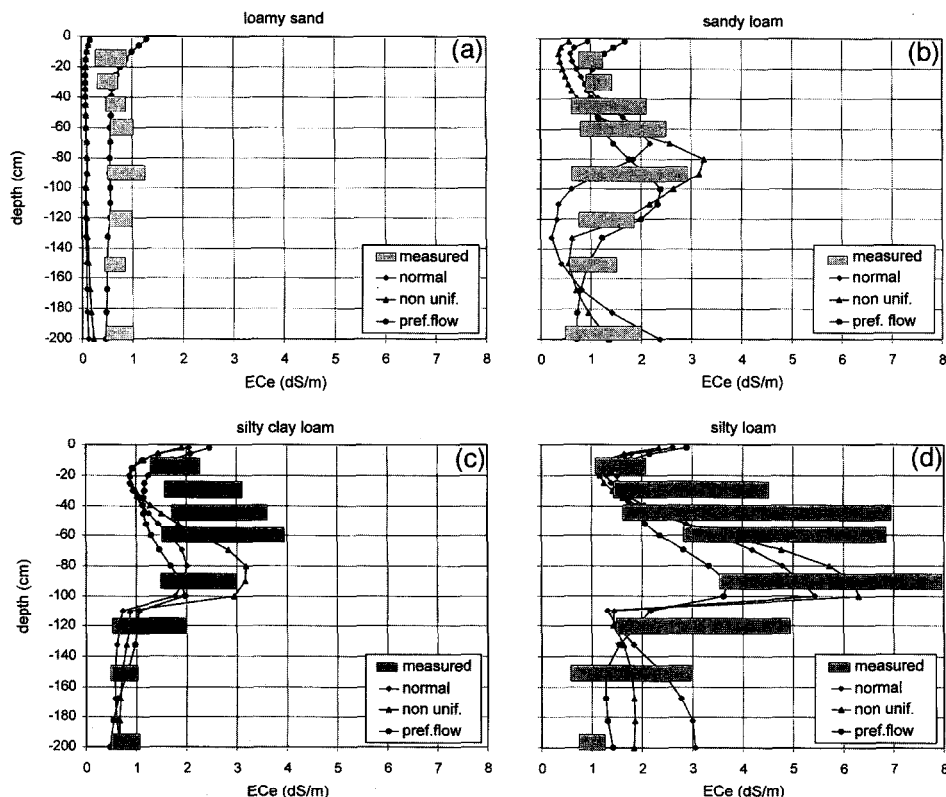


Fig. 3. Measured and simulated EC_e profiles with normal, non-uniform and preferential flow concept on four soils.

The non-uniform concept is clear and practical and the results corresponds very well with field observations. However, the division of the field in several sub-units, receiving a certain percentage of the irrigation water is very arbitrary. To justify this division, a detailed study of the infiltration pattern over the field should be carried out. Secondly, the non-uniform concept is laborious, since three simulations have to be done and processed instead of only one.

It is preferable to use the 'normal' calibration, even though the heterogeneity of water infiltration over the field is not taken into account. The 'normal' calibration is straight, simple, and yields acceptable results and is not time consuming. This concept was validated by the water balance for the period July–December 1995. Both simulated pressure heads and moisture content matched well the measured values. This model was, therefore, used in the scenario-analysis.

3.3. Sensitivity analysis

A sensitivity analysis for the silty loam (field 4) was carried out in order to identify the most sensitive input parameters and recommend data collection in the field and/or

Table 4

Results (RT and SSC, see Eqs. (1) and (2)) of sensitivity analysis for calibrated model of a silty loam soil

Scenario	Relative transpiration (RT)	Salt storage change (SSC)
Reference	0.93	0.14
Decrease rooting depth (with 50%)	0.89	0.10
Decrease crop factors (with 25%)	0.99	– 0.48
Decrease Boesten parameter (with 33%)	0.95	– 0.15
Decrease saturated moisture content (with 16%)	0.91	– 0.02

laboratory. The calibration period was taken as a reference scenario for the sensitivity analysis, while changing rooting depth and distribution, crop factors, Boesten parameter for actual soil evaporation, and saturated moisture content.

Table 4 shows the most important results of the sensitivity analysis. The relative transpiration is affected by the rooting depth of the crop, and should therefore be measured in the field. Root water extraction is more effective with deeper roots, since less water is lost by percolation, especially on permeable soils. In case of shallow roots, percolation is increased and more salts are leached. The distribution of roots is not significantly affecting the transfer of water and salts, and can be adapted from literature.

Decreasing crop factors results in a lower potential evapotranspiration, so the water requirements of the crop are met, and the excessive irrigation leaches the salts. The values of the crop factors strongly affect the actual transpiration and thus, the percolation. A decrease in crop factors by about 10%, might double the percolation. Therefore, preferably local crop factors should be used instead of general values as supplied by Doorenbos and Pruitt (1977).

Decreasing the Boesten parameters results in a considerable decrease in the actual soil evaporation. Thus, in the upper soil layers more water is available for the crop and the percolation of water and salts is increased. Boesten parameters can be derived from literature or, if available, microlysimeter experiments (Boast and Robertson, 1982).

The wetting front moves faster through a soil with a smaller saturated moisture content, thus, decreasing the relative transpiration and increasing the leaching of salts. It is recommended to measure the saturated moisture content in the laboratory and the saturated hydraulic conductivity in the field. The remaining shape parameters can be derived from literature (Wösten et al., 1987) and, if needed, adapted during the calibration process.

4. Results and discussion

4.1. Short term scenario analysis

4.1.1. Management variables

Several irrigation management scenarios are simulated and compared with the simulation results of a reference irrigation scenario, which is based on the recommended irrigation practices of the authorized Agricultural Department (see Table 5). It is

Table 5

Description of management scenarios in terms of quantity (cm), quality (dS/m) and frequency (-) of applications

Scenario	Wheat season						Cotton season					
	Quantity		Quality		Frequency		Quantity		Quality		Frequency	
	Pre	Other	Pre	Other	Pre	Other	Pre	Other	Pre	Other	Pre	Other
Reference	15.0	7.5	0.2	1.5	1	5	10.0	7.8	0.2	1.5	2	8
Under-irrigation	10.0	5.6	0.2	1.5	1	5	7.5	5.9	0.2	1.5	2	8
Low frequency	15.0	12.5	0.2	1.5	1	3	10.0	10.4	0.2	1.5	2	6
Pre-sowing	15.0	7.5	1.5	1.5	1	5	10.0	7.8	1.5	1.5	2	8
Tube well	15.0	7.5	3.0	3.0	1	5	10.0	7.8	3.0	3.0	2	8

recommended to apply a pre-sowing irrigation with canal water ($EC_{iw} = 0.2$ dS/m) to leach the salts in the beginning of the growing season and to wet the profile. In the under-irrigation scenario, the irrigation depths are decreased to 75% of the reference depths. In the low frequency scenario the number of gifts is reduced, while the total amount of irrigation water is kept the same as that of the reference scenario. In the pre-sowing scenario, there is no irrigation with canal water at the beginning of the growing season, but with water of an inferior quality. The tube well scenario simulates a situation where a farmer has access to poor quality tube well water only. The results of the scenario-analysis are listed in Table 6. The initial salt storage of all scenario simulations was set at around 1000 g/m² in order to compare the obtained results.

In the reference scenario, the wheat crop is over-irrigated to a larger extent than the cotton crop, which is evident from the higher salts storage change over the cotton season. For the under-irrigation scenario, the water requirements of the wheat crop are still met, but there is hardly any percolation and salts build up in the soil. During the cotton season, a considerable reduction in crop transpiration takes place and the salinization process continues (see Fig. 4). In the low frequency scenario, the larger irrigation gifts saturate the soil and the wetting front moves faster through the soil profile, so less water is extracted by the roots and more salts are leached. This phenomenon is not very pronounced on the silty loam (see Table 6), but is strongly present on permeable sandy soils with a smaller retention capacity. Managing the water and salt balance by means of changing the irrigation interval is only effective in case

Table 6

Results (RT and SSC, see Eqs. (1) and (2)) of scenario analysis of management variables for a silty loam soil

Scenario	Wheat season		Cotton season		Total agricultural year	
	RT	SSC	RT	SSC	RT	SSC
Reference	1.00	0.00	0.99	0.24	1	0.20
Under-irrigation	0.98	0.26	0.85	0.37	0.89	0.73
Low frequency	0.99	-0.22	0.98	0.55	0.99	0.18
Pre-sowing	1.00	0.12	0.98	0.33	0.99	0.44
Tube well	0.99	0.61	0.93	0.67	0.95	1.66

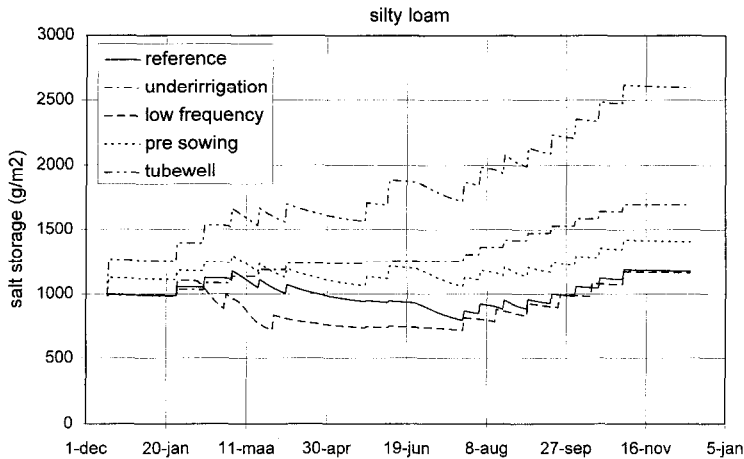


Fig. 4. Development of salt storage on a silty loam over one wheat–cotton cycle for different management scenarios.

there is a percolation component. In case of under-irrigation, when all the applied irrigation water is extracted by the roots, there is no leaching of salts. Thus, the build up of salts in the soil is equal to the amount of salts applied with the irrigation water, regardless of the frequency of irrigation.

Fig. 4 illustrates the impact of a pre-sowing irrigation with poor quality water instead of canal water on the leaching of salts in the beginning of the growing season. The salt storage is calculated up to a depth of 210 cm. A pre-sowing irrigation with canal water is highly preferable, since the germination stage is the most sensitive period of a crop with respect to salinity. The tube well scenario shows the disastrous effect of poor quality tube well water on soil salinization (Fig. 4). The higher electrical conductivity of the irrigation water is significantly reducing the crop transpiration, especially for the less over-irrigated cotton crop.

4.1.2. Environmental parameters

The impact of two important environmental parameters is studied: soil type and groundwater table depth. The reference and under-irrigation scenarios are performed for the four selected soil types, in case of a field with a deep groundwater table (free drainage) and a field with a groundwater table at 2 m depth. The EC of the groundwater is 1.5 dS/m, which is equal to the EC of the irrigation water, except for the pre-sowing irrigations with canal water. The results are shown in Table 7.

In case of a deeply drained field, the percolation on sandy soils (LS and SL) is much higher than on loamy soils (SiCL and SiL) due to higher conductivity and lower retention capacity. Consequently, less water is extracted by roots on sandy soils compared to loamy soils and the leaching of salts is more effective. For the under-irrigation scenario, the same trends can be observed as for the reference scenario, but even more pronounced. For the loamy soils, almost all the applied irrigation water is

Table 7

Results (RT and SSC, see Eqs. (1) and (2)) of scenario analysis of environmental parameters for reference and under-irrigation management scenario

Soil type	Deep groundwater table				Groundwater table at 2 m depth			
	Reference		Under-irrigation		Reference		Under-irrigation	
	RT	SSC	RT	SSC	RT	SSC	RT	SSC
Loamy sand (LS)	0.92	−0.27	0.78	−0.06	0.94	0.30	0.84	0.63
Sandy loam (SL)	0.93	−0.17	0.80	0.20	0.95	0.34	0.85	0.65
Silty clay loam (SiCL)	0.98	0.09	0.84	0.74	1.00	0.58	0.93	0.76
Silty loam (SiL)	1.00	0.20	0.89	0.74	1.00	0.59	0.96	0.90

transpired by the crop, so there is hardly any percolation and the build up of salts in the soil equals the amount of salts applied through irrigation.

In case a shallow groundwater table is present, capillary rise will contribute to an increase in crop transpiration. When a field is under-irrigated capillary water supply increases the relative transpiration considerably, especially for the loamy soils. At the same time, a shallow groundwater table causes a higher salt storage in the soil. This can be explained by the increased wetness of the profile, resulting in lower salt concentrations and thus, a smaller amount of salt is leached with the percolating water. In case of

Table 8

Results (RT and SSC, see Eqs. (1) and (2)) of long term analysis of current irrigation practices with various irrigation water quantities (D_{iw} in cm) and various water qualities (EC_{iw} in dS/m) for four soils

Scenarios		Loamy sand		Sandy loam		Silty clay loam		Silty loam	
D_{iw}	EC_{iw}	RT	SSC	RT	SSC	RT	SSC	RT	SSC
133	1	0.92	−0.43	0.93	−0.33	0.98	−0.07	1.00	0.07
120	1	0.88	−0.40	0.89	−0.19	0.95	0.43	0.96	1.03
107	1	0.83	−0.31	0.84	0.13	0.88	2.20	0.97	3.42
93	1	0.75	−0.07	0.77	1.01	0.77	4.51	0.75	5.72
80	1	0.66	0.45	0.76	2.69	0.64	6.75	0.62	8.05
133	2	0.89	0.12	0.92	0.34	0.96	0.84	0.97	1.09
120	2	0.85	0.18	0.88	0.60	0.93	1.57	0.93	2.26
107	2	0.80	0.29	0.82	1.14	0.85	3.28	0.84	4.33
93	2	0.72	0.56	0.72	2.34	0.74	5.45	0.73	6.58
80	2	0.63	1.05	0.64	4.12	0.62	7.71	0.61	9.02
133	3	0.86	0.64	0.90	1.00	0.94	1.71	0.95	2.05
120	3	0.82	0.72	0.86	1.35	0.89	2.56	0.90	3.20
107	3	0.76	0.84	0.82	1.99	0.82	4.16	0.81	5.10
93	3	0.69	1.07	0.72	3.21	0.71	6.22	0.70	7.31
80	3	0.61	1.48	0.62	4.95	0.60	8.46	0.59	9.78
133	4	0.83	1.09	0.87	1.54	0.91	2.43	0.92	2.81
120	4	0.78	1.20	0.83	1.69	0.86	2.89	0.86	3.47
107	4	0.73	1.33	0.77	1.97	0.79	3.68	0.78	4.40
93	4	0.66	1.52	0.69	2.48	0.69	4.58	0.68	5.37
80	4	0.60	1.86	0.60	3.12	0.58	5.28	0.57	6.10

under-irrigation, salts brought into the root zone through capillary rise contribute to the soil salinity as well.

4.2. Long term scenario analysis

Long term scenarios (10 yr) are performed in order to predict trends in soil salinity and crop transpiration under current irrigation practices. The scenarios are based on the previously discussed reference scenario, changing the irrigation depths and the irrigation

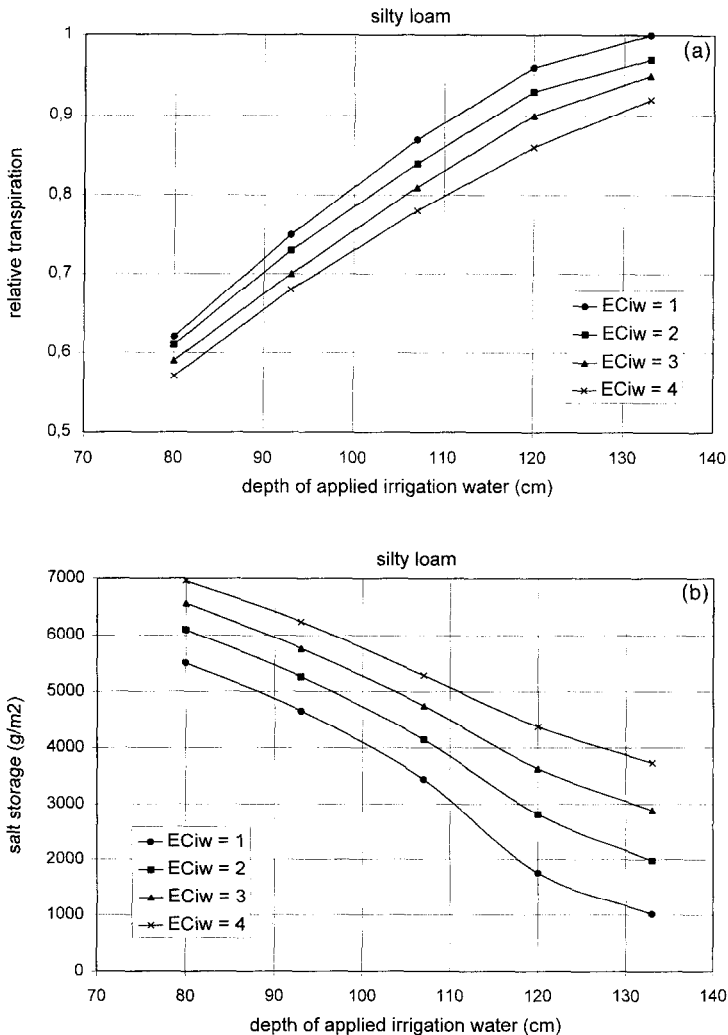


Fig. 5. Relative transpiration and salt storage on a silty loam for long term under-irrigation scenarios with different water qualities.

water quality, while keeping all other variables constant (e.g. irrigation interval). Often, farmers do not have enough irrigation water at their disposal and crop water requirements are not fully met. This stresses the importance of studying the impact of under-irrigation on salinization and crop transpiration. Therefore, five under-irrigation scenarios are performed, decreasing the application depths up to 60% of the reference application depths (10% interval). As farmers are using more and more tube well water for irrigation, the irrigation water quality of the scenarios is varied from 1 to 4 dS/m (interval 1 dS/m).

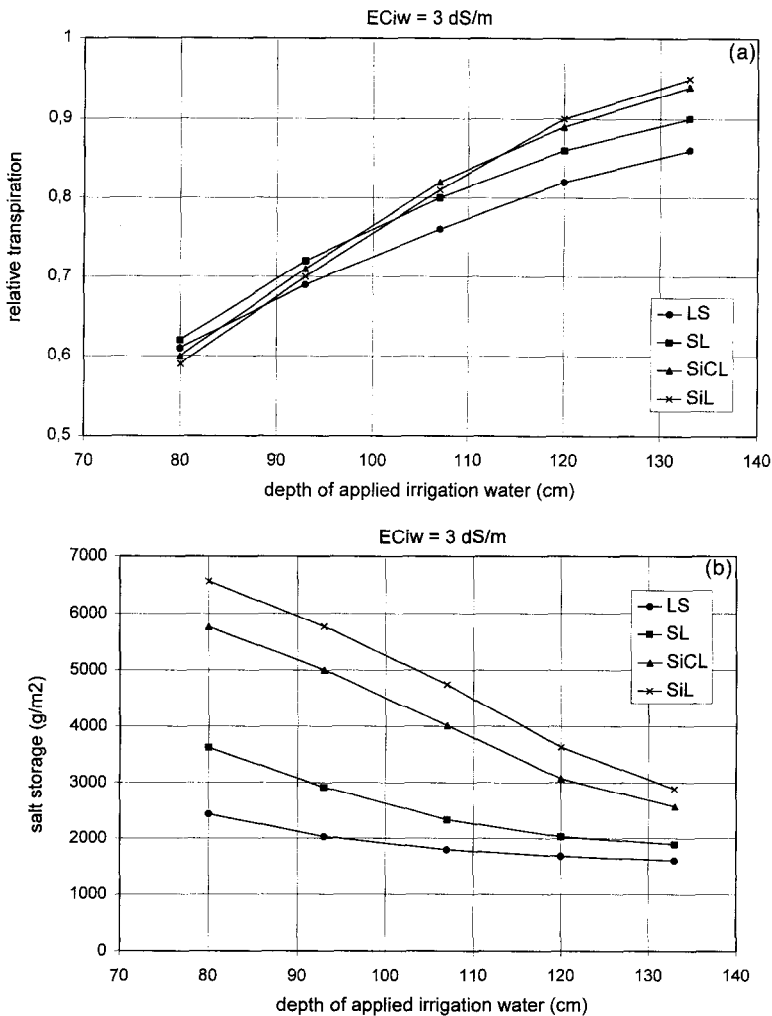


Fig. 6. Relative transpiration and salt storage on four soils for long term under-irrigation scenarios with $EC_{iw} = 3 \text{ dS/m}$.

After a few years, the salinization process reaches a certain equilibrium state with a constant salt storage: the amount of salt added to the soil is equal to the amount of salt leached from the soil. The equilibrium state is reached earlier on sandy soils (after two years) than on loamy soils (after four years). In Table 8 the results of the scenario analysis are listed (equilibria).

Table 8 shows that the relative transpiration is mainly affected by the applied irrigation depth, and less influenced by the electrical conductivity of the irrigation water. Fig. 5 illustrates that the relative transpiration is increasingly reduced with under-irrigation, while salts accumulate in the soil profile. The decrease in relative transpiration due to a poor water quality of the irrigation water (from 1 to 4 dS/m) is approximately 8%, while the reduction in relative transpiration due to under-irrigation (from 133 to 80 cm) is 38% on loamy soil. The build up of salts in the soil (calculated up to a depth of 210 cm) increases with a higher electrical conductivity of the irrigation water, but is mostly affected by the extent of under-irrigation.

The salinization process takes place differently on sandy and loamy soils, which is shown in Fig. 6. Due to a lower conductivity, a higher retention capacity and consequently a higher root water uptake, loamy soils are more susceptible to salinization, even for 10% under-irrigation ($D_{iw} = 120$ cm). On sandy soils, salinization is not severe, even for 40% under-irrigation ($D_{iw} = 80$ cm), as sufficient leaching of salts still occurs. Generally, crop transpiration is higher on loamy soil than on sandy soil. Fig. 6, however, shows that crop transpiration on sandy soils is higher than loamy soils for an applied irrigation depth of 80 cm. This can be explained by the increased salt storage of loamy soils at this extent of under-irrigation. The soil salinity of the 210 cm profile is considerably affecting the root water extraction, and thereby dominating the reduction in transpiration due to insufficient irrigation application.

Based on the model output, an indicative risk analysis can be carried out for the current irrigation practices in the Chishtian subdivision. For a certain irrigation water quantity (D_{iw}) and a certain irrigation water quality (EC_{iw}), the reduction in crop transpiration on a certain soil type can be estimated and evaluated.

5. Conclusions

This study evaluated the impact of an increasing use of groundwater for irrigation on soil salinization and crop transpiration in the command area of an irrigation system in Punjab, Pakistan. The approach involved a scenario analysis with the agro-hydrological model SWAP93 for four different soil types for cotton and wheat. Simulation of soil salinity with a one-dimensional model has to deal with the problem of non-uniform water distribution over a field, resulting in a higher average soil salinity over a field than simulated with the model. Although model concepts were developed to tackle this problem, it is preferable to use SWAP93 in its original and simple way, as satisfactory results were still obtained. Although it is preferable to determine input parameters, particularly those related to the soil hydraulic functions, independently in situ, it was shown that satisfactory calibration of the model is possible by using the standard soils series as a starting point. Other input parameters can generally be obtained from the

literature if reliable field data are not available. The sensitivity analysis showed that accurate data on crop factors, soil evaporation, rooting depth and soil hydraulic functions are needed to attain reliable water and salt balances. The advantage of using a relatively complex model like SWAP93 is that by making a large number of relationships explicit in the model, the relative impact of a large number of parameters on water and salt balances can be determined, once the model is calibrated and validated.

A scenario-analysis for management variables (e.g., quantity, quality and frequency of irrigation) and environmental parameters (e.g., soil type and groundwater table depth) was performed. The interval of irrigation appeared to be a management option for farmers to control their salinity and optimize their crop transpiration. By irrigating sandy soils less frequently with larger application depths, crop transpiration is reduced, but, at the same time, salinity is kept at acceptable levels. Another way to keep salinity levels at an acceptable level is to apply a pre-sowing irrigation with canal water to leach the salts at the beginning of the growing season.

A long term analysis was carried out for the current irrigation practices in the study area. Soil texture considerably affects the water and salt balances. Up to a certain point, crop transpiration is more reduced on sandy soils than on loamy soils when decreasing the irrigation quantity. However, if the extent of under-irrigation is increased, crop transpiration becomes higher on sandy soils than on loamy soils due to salinity stress which is present on loamy soils but not on the permeable sandy soils. Reducing the irrigation quantity is far more affecting the crop transpiration and salinization process than the quality of the applied water when looking at the total concentration of salts. However, the pumped groundwater in the area contains a relatively high amount of sodium and there is danger of sodification. Therefore, the utilization of a solute transport model like SWAP93 should be undertaken preferably in conjunction with a study of the geo-chemical processes.

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